

Camera Calibration and Real-World Measurement

Module 2: Steps 1–3 and Theory

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Code repository

https://github.com/RagDoll95/CSC8830_Assignments

All code, captured images and logged measurements are in that repository. The Module 2 code is under `week_2/module2/` and the web application that demonstrates it is at the repository root under `app/`, started with `week_2/.venv/bin/python app/app.py`. Its three Module 2 pages cover the calibration, the measurement itself (click two points on a photo taken at a known distance) and the logged measurements with their error statistics.

1 Step 1: Camera calibration

A smartphone camera was calibrated with OpenCV’s built-in tools by [Zhang’s multi-plane method](#). A 9×6 inner-corner checkerboard with a measured square size of 24.9 mm was photographed 24 times in varied orientations. Corners were located and refined to sub-pixel accuracy, and the camera model was then fitted with `cv2.calibrateCamera`, which minimises total reprojection error over the intrinsics, the distortion coefficients and the per-view pose. Focus, exposure and zoom were locked for the whole set, and every image was captured at the one resolution later used in Steps 2 and 3.

The recovered camera matrix and distortion coefficients are

$$K = \begin{bmatrix} 3336.69 & 0 & 1496.68 \\ 0 & 3335.58 & 2020.27 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 0.2231 \\ -1.3758 \\ 0.00058 \\ -0.00081 \\ 2.7090 \end{bmatrix}, \quad (1)$$

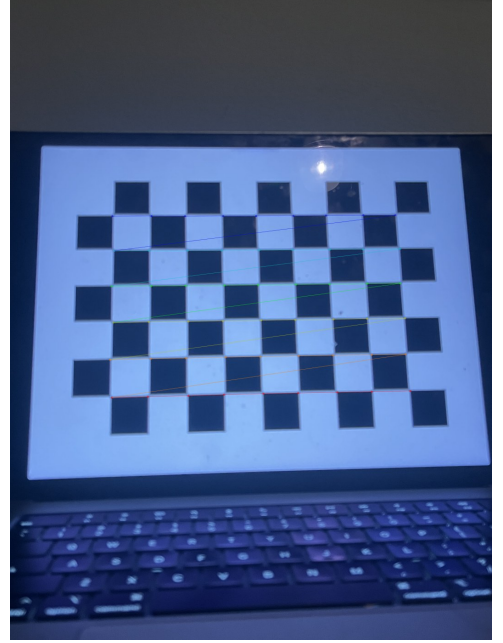
in the ordering $(k_1, k_2, p_1, p_2, k_3)$. Since f and c are in pixels, K is valid only at the 3024×4032 resolution it was fitted at. Table 1 collects the fit and Figure 1 shows the detected corners on the best- and worst-fitting views.

Table 1: Calibration summary.

Quantity	Value
Images used	24
Pattern (inner corners)	9×6
Square size	24.9 mm
Resolution	3024×4032
f_x	3336.69 px
f_y	3335.58 px
c_x	1496.68 px
c_y	2020.27 px
f_y/f_x	0.99967
Overall RMS reprojection error	1.863 px
Median per-image RMS	0.581 px



(a) IMG_3913, RMS 0.364 px.



(b) IMG_3921, RMS 7.475 px.

Figure 1: Detected corner overlays for the two of the 24 calibration views

2 Step 2: Real-world dimensions from perspective projection

Under the pinhole model a point $P = (X, Y, Z)$ in camera coordinates projects to

$$u = f_x \frac{X}{Z} + c_x, \quad v = f_y \frac{Y}{Z} + c_y. \quad (2)$$

A single image cannot recover P , since every point on the ray through (u, v) maps to the same pixel. Supplying the object distance Z independently removes that one unknown and makes Equation 2 invertible:

$$X = (u - c_x) \frac{Z}{f_x}, \quad Y = (v - c_y) \frac{Z}{f_y}. \quad (3)$$

A dimension between two selected points p_1 and p_2 lying on a common plane at depth Z is then the Euclidean distance between their backprojections,

$$L = \|P_2 - P_1\| = Z \sqrt{\left(\frac{\Delta u}{f_x}\right)^2 + \left(\frac{\Delta v}{f_y}\right)^2}, \quad (4)$$

which for each axis comes out to:

$$W = \Delta u \frac{Z}{f_x}, \quad H = \Delta v \frac{Z}{f_y}. \quad (5)$$

Width, height and diagonal therefore all come from Equation 4 by one code path. Both endpoints must lie at the same depth Z , and lens distortion is removed from the two selected points only, with `cv2.undistortPoints`, which returns *normalised* coordinates, already X/Z and Y/Z , so the dimension is $Z \cdot \|\Delta(\text{normalised})\|$ with no further division by f_x .

Figure 2 Shows an example. The two selected points span 1069.7 px across the top edge of a laptop lid at a tape-measured $Z = 1000$ mm, with a checkerboard displayed on its screen, and Equation 4 turns that span into 317.7 mm against a ruler truth of 356.0 mm. It is entry 1 of the validation set in Step 3.

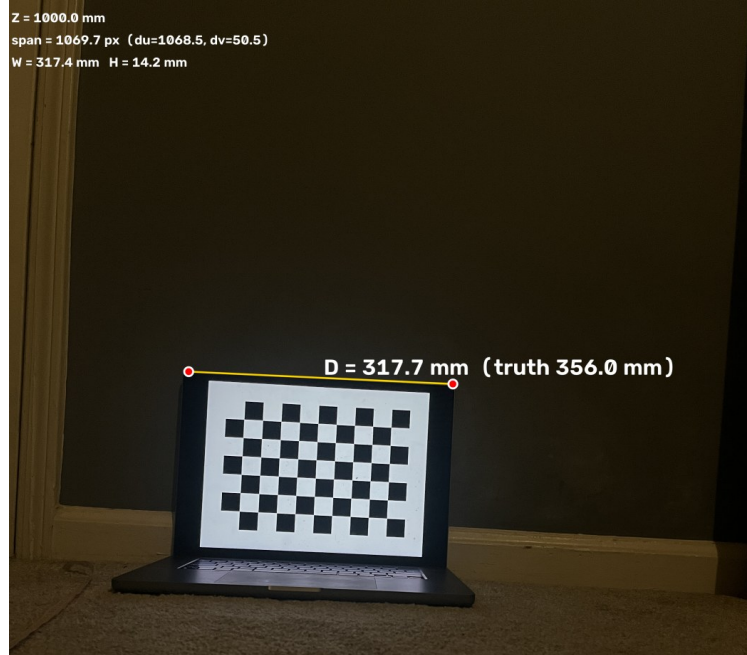
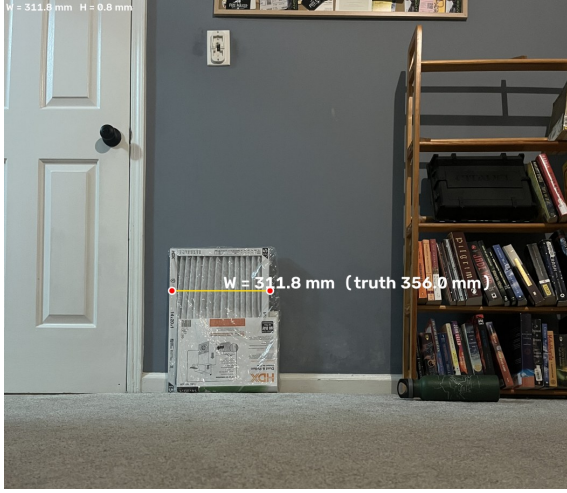


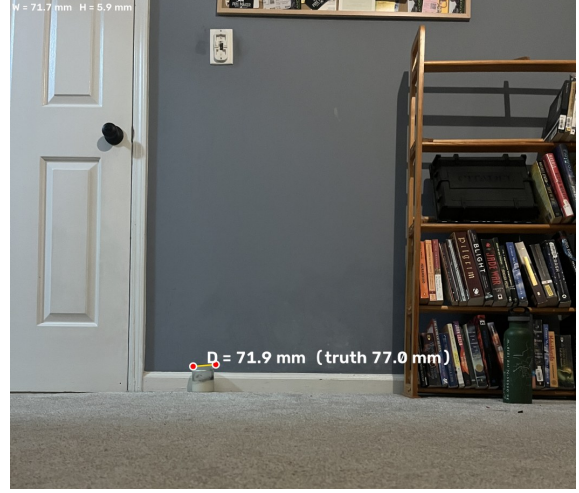
Figure 2: Step 2 on a single capture at $Z = 1000$ mm: the two selected points, the pixel span between them, and the dimension Equation 4 returns for it.

3 Step 3: Validation experiment

Twenty object measurements were validated against ruler ground truth. Ground truth was recorded *before* imaging. Both the object and the measured dimension were varied: 15 diagonals and 5 widths, over ground-truth sizes from 77 to 413 mm. Nineteen measurements were taken at a nominal $Z = 2000$ mm, satisfying the requirement that the object be at least 2 metres from the camera, and one at $Z = 1000$ mm. Each row stores the two full-resolution pixel coordinates alongside the result, so every logged value can be re-derived from its pixels; doing so reproduces the table to within 0.005 mm. Figure 3 shows two of the annotated measurements, Table 2 lists all twenty, and Table 3 and Figure 4 report the error statistics over them.



(a) Air filter, W : measured 311.8 mm against 356.0 mm.



(b) Candle, D : measured 71.9 mm against 77.0 mm.

Figure 3: Two of the 20 recorded measurements, at opposite ends of the size range.

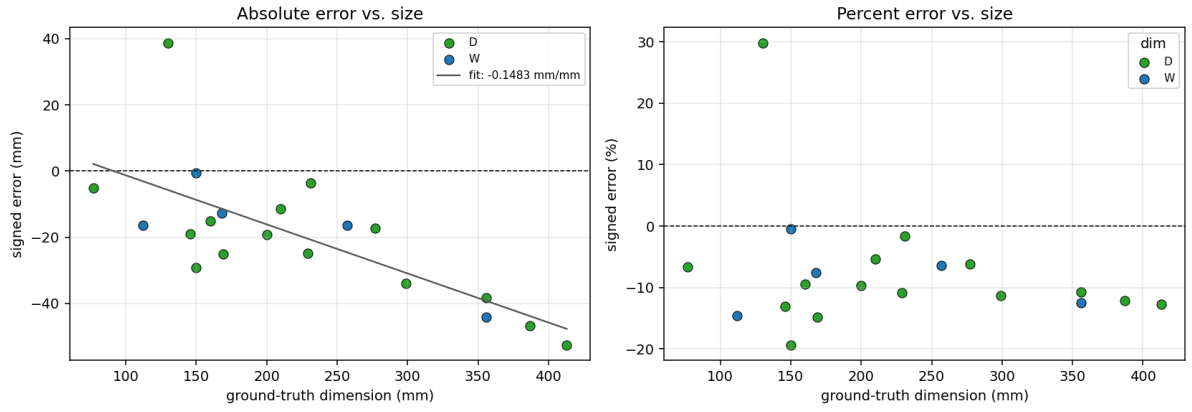
Table 2: The 20 validation measurements. Error is measured – truth.

id	object	dim	Z (mm)	truth (mm)	meas (mm)	err (mm)	err (%)
1	Mac	D	1000	356.0	317.7	-38.3	-10.76
2	Paperback book	D	2000	210.0	198.7	-11.3	-5.40
3	Water bottle	D	2000	277.0	259.9	-17.1	-6.19
4	Hydroflask	D	2000	130.0	168.8	+38.8	+29.84
5	Headphone case	W	2000	150.0	149.4	-0.6	-0.41
7	Game controller	D	2000	150.0	120.9	-29.1	-19.41
8	clothes hangar	D	2000	413.0	360.3	-52.7	-12.75
9	Hardback book	D	2000	160.0	144.9	-15.1	-9.45
10	Tissue box	D	2000	231.0	227.3	-3.7	-1.60
11	Candle	D	2000	77.0	71.9	-5.1	-6.58
12	clothing iron	D	2000	299.0	265.1	-33.9	-11.32
13	Climbing shoes	D	2000	229.0	204.2	-24.8	-10.82
14	Sketchbook	W	2000	257.0	240.6	-16.4	-6.38
15	Duct Tape	D	2000	146.0	127.0	-19.0	-13.02
16	Glasses case	D	2000	169.0	143.9	-25.1	-14.82
18	Power supply unit	W	2000	168.0	155.3	-12.7	-7.54
19	Air filter	W	2000	356.0	311.8	-44.2	-12.42
20	Mouse	W	2000	112.0	95.6	-16.4	-14.63
21	tennis racket	D	2000	387.0	340.2	-46.8	-12.08
22	paperback book 2	D	2000	200.0	180.7	-19.3	-9.66

Table 3: Error statistics over the 20 measurements.

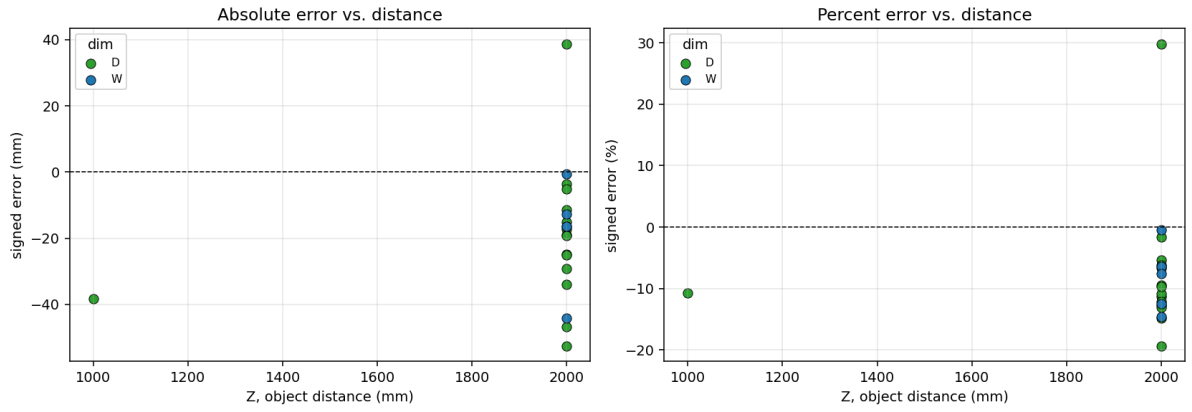
Statistic	Value
Measurements, n	20
Ground-truth size range	77.0–413.0 mm
Distance range	1000–2000 mm
Mean signed error (bias)	−19.636 mm
Standard deviation of error	19.914 mm
MAE	23.515 mm
RMSE	27.610 mm
Mean signed percent error	−7.771%
MAPE	10.755%
Std. deviation of percent error	9.948%
Most negative error	−52.670 mm (id 8)
Most positive error	+38.790 mm (id 4)
Standard error of the mean	4.453 mm
95% CI on mean signed error	[−28.364, −10.908] mm
corr(size, signed error)	−0.721
corr(Z , percent error)	+0.071

Step 3: error vs. ground-truth dimension (n=20)

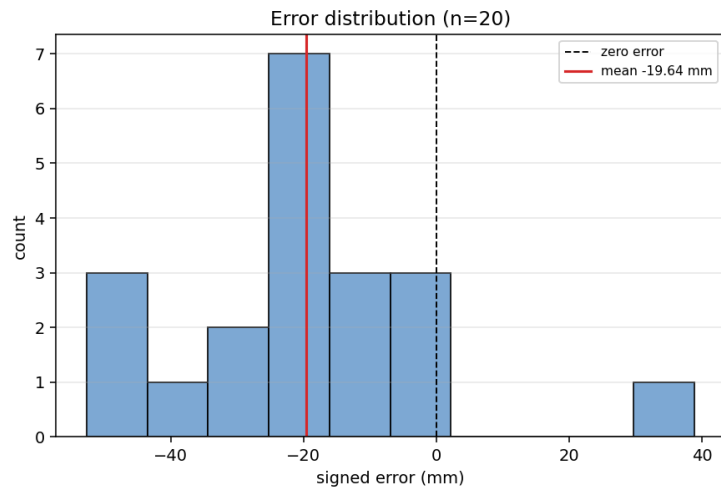


(a) Error against ground-truth size, in millimetres and in percent.

Step 3: error vs. distance (n=20)



(b) Error against object distance; all but one measurement was taken at $Z = 2000$ mm.



(c) Distribution of percent error.

Figure 4: Error statistics of the 20 validation measurements.

4 Theory: relating the image of a point in two cameras

Camera 1 is static; camera 2 sits at some distance from it and at an oblique orientation. A point $P = (X, Y, Z)$ lies in the field of view of both. What follows relates x_1 , the image of P in camera 1, to x_2 , its image in camera 2. Once the assumptions below are in place, the relationship is one substitution.

4.1 Assumptions

1. **Ideal pinhole projection.** Both images are undistorted first, using the distortion coefficients from Step 1. After correction the residual departure from an ideal pinhole is below the corner-localisation noise, so modelling it further would add parameters without reducing error.
2. **Both cameras are calibrated.** K_1 and K_2 are known from a Zhang calibration as in Step 1. If one phone captures both views, as it does here, then $K_1 = K_2 = K$ and half the unknown intrinsics disappear.
3. **Zero skew, square pixels.** The Step 1 fit gives f_y/f_x within 0.03% of unity and CMOS pixel axes are orthogonal by construction, so the skew term in K is set to zero rather than estimated.
4. **Rigid relative pose.** R and $\mathbf{t}$ are constant for the pair of exposures. With one phone moved by hand they describe those two exposures and nothing else.
5. **World frame anchored to camera 1.** Camera 1's extrinsics become $[I \mid \mathbf{0}]$, so P in the world frame is also P in camera 1's frame.

4.2 The derivation

With the world frame on camera 1, the two projections are

$$x_1 = K_1 P, \quad x_2 = K_2 (R P + \mathbf{t}), \quad (6)$$

where $x_1 = (u_1, v_1, 1)^\top$ and $x_2 = (u_2, v_2, 1)^\top$ are the image points, R is camera 2's rotation relative to camera 1, a proper 3D rotation with $R^{-1} = R^\top$, and $\mathbf{t} = -R\tilde{C}$ with $\tilde{C}$ the centre of camera 2 expressed in camera 1's frame. "Oblique orientation" is the statement $R \neq I$, and "at some distance" is $\mathbf{t} \neq \mathbf{0}$.

Only the first of the two has to be inverted, and Step 2 already did it. The intrinsic matrix is upper triangular, so its inverse is immediate,

$$K_1 = \begin{bmatrix} f_{x1} & 0 & c_{x1} \\ 0 & f_{y1} & c_{y1} \\ 0 & 0 & 1 \end{bmatrix}, \quad K_1^{-1} = \begin{bmatrix} 1/f_{x1} & 0 & -c_{x1}/f_{x1} \\ 0 & 1/f_{y1} & -c_{y1}/f_{y1} \\ 0 & 0 & 1 \end{bmatrix}, \quad (7)$$

and applying it to x_1 strips the intrinsics off the measured pixel:

$$K_1^{-1} x_1 = \begin{pmatrix} (u_1 - c_{x1})/f_{x1} \\ (v_1 - c_{y1})/f_{y1} \\ 1 \end{pmatrix} = \begin{pmatrix} X/Z \\ Y/Z \\ 1 \end{pmatrix}, \quad \text{so} \quad P = Z K_1^{-1} x_1. \quad (8)$$

The middle step is Equation 3 written as a vector: x_1 fixes the direction of the ray on which P lies, and the scalar Z says how far along that ray P sits. Substituting this P into camera 2's projection is then the whole derivation,

$$\begin{aligned}
x_2 &= K_2(RP + \mathbf{t}) \\
&= K_2(R(Z K_1^{-1}x_1) + \mathbf{t}) && \text{(substitute } P) \\
&= Z K_2 R K_1^{-1}x_1 + K_2\mathbf{t} && (R, K_2 \text{ are linear; } Z \text{ is a scalar}) \\
&= Z \left(K_2 R K_1^{-1}x_1 + \frac{1}{Z} K_2\mathbf{t} \right) && \text{(factor out } Z).
\end{aligned} \tag{9}$$

That leading Z multiplies every entry of the bracket, including the third, so it divides straight back out when the pixels are read off and can be dropped. What is left is the relationship between the two images of P :

$$\boxed{x_2 = K_2 R K_1^{-1}x_1 + \frac{1}{Z} K_2\mathbf{t}} \tag{10}$$

The image of P in camera 2 is therefore x_1 warped by the rotation and the two sets of intrinsics, then displaced by an amount inversely proportional to the depth of P . That depth is the only per-point quantity in the relation; everything else is fixed for the pair of cameras.

4.3 Parameters and variables

Table 4: The static parameters and the per-point variables in Equation (10), and how each is obtained.

Symbol	Type	Meaning	How obtained
K_1, K_2	static	Intrinsic matrix of each camera	Zhang calibration, Step 1. Equal if one phone takes both views.
R	static	Relative rotation, camera 1 $\rightarrow$ camera 2	<code>cv2.stereoCalibrate</code> with the board visible in both views, or from point correspondences.
$\mathbf{t}$	static	Relative translation, $\mathbf{t} = -R\tilde{C}$	As for R . From correspondences alone it is direction-only; metric scale needs one known length.
x_1, x_2	variable	Image coordinates of P in each view	Measured: feature detection and matching, or manual selection.
$P = (X, Y, Z)$	variable	The scene point	Unknown; recovered by triangulating x_1 and x_2 once R and $\mathbf{t}$ are known.
Z	variable	Depth of P in camera 1's frame	Triangulated from the two views, or measured with a tape as in Step 3.